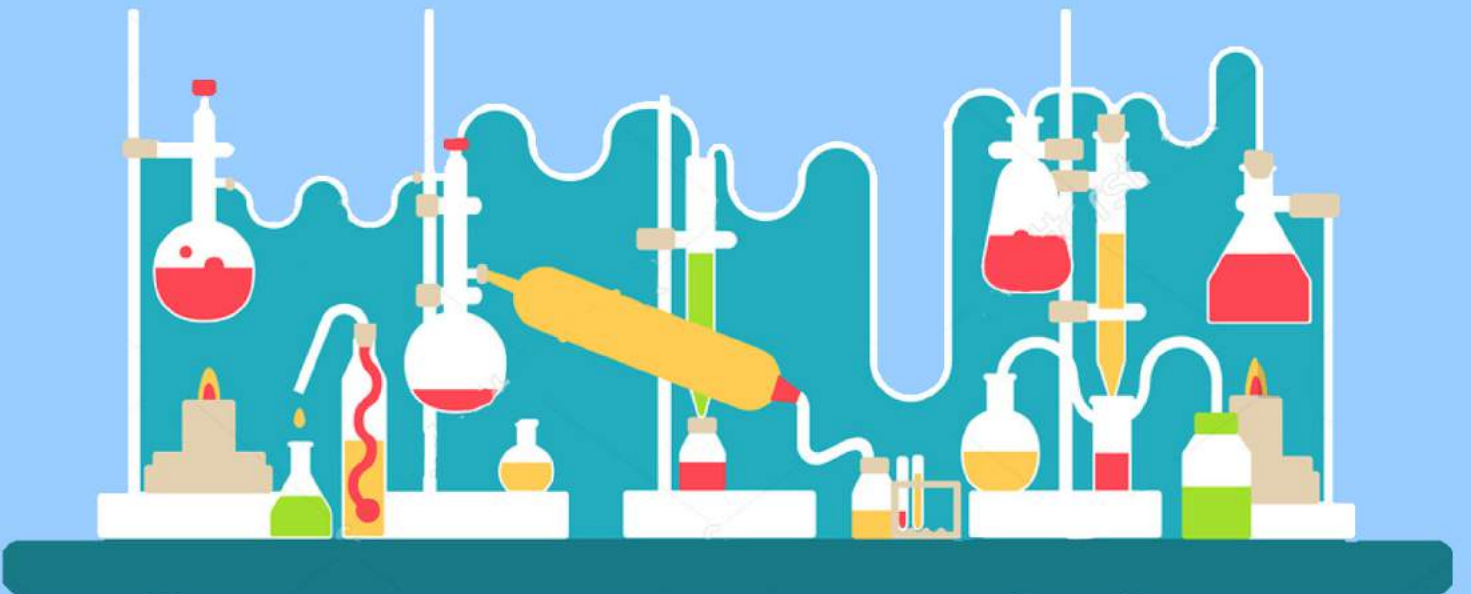




BIOLOGY LABORATORY



#2 Cell Structure



نضع بين أيديكم هذا العمل الذي كان نتاجًا للتعاون بين (لجنة الطبّ البشريّ) و (لجنة طبّ الأسنان)، و نُوجِّهه لطلاب السّنة الأولى (دُفعة 2016) كمصدرٍ رئيسيّ و شاملٍ لمادّة (مختبر البيولوجي) ، و نضمّن و فيه شرحً لكم اشتماله على المطلوب منكم من الكتاب و السلايدات، موضّح للأفكار المهمّة، و قد بذلنا جهدنا أن نُخرجه لكم بالرجوع للمصادر الموثوقة، هذا و قد تجدون بعض التوضيحات باللّغة العربيّة 3: تيسيرًا عليكم؛ إذ هذا أدعى لترسيخ الفهم

تعاون زملائكم على إخراج بصورته المُتقنة إذ أدركوا مسعى الإتيان و مقصده، فكان أن وضعوا أمام أعينهم صورة الطّبيب الذي ستكونوه كلّكم، فلم يرضوا إلّا بأن يكون طبيبًا حقًا، تقديرًا منّا للعلم و أهله، و إيمانًا بأنّ بذرة الخير لا تُتْبِتُ إلّا خيرًا :"

كلّ التوفيق ٨٨"

Modern cell theory principles:

1. All known living things are made up of one or more cells.
2. All living cells arise from pre-existing cells by division.
3. The cell is the fundamental unit of structure and function in all living organisms.
4. The activity of an organism depends on the total activity of independent cells.
5. Energy flow (metabolism and biochemistry) occurs within cells.
6. Cells contain DNA which is found specifically in the chromosome and RNA found in the cell nucleus and cytoplasm.
7. All cells are basically the same in chemical composition in organisms of similar species.

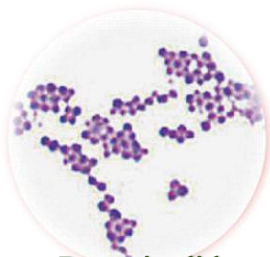
Cell الخلية	The basic structural & functional units of life. They can be either prokaryotic or eukaryotic.
Multicellular organisms كائنات عديدة الخلايا	Like plants & animals, they are a collection of many kinds of specialized cells (خلايا متخصصة) that could not survive for a long time on their own. لا تنجو من تلقاء نفسها فهي تحتاج إلى غذاء.
Common features for cells الخصائص المشتركة للخلايا	All cells share certain basic features: 1- They are all bounded by plasma membrane. 2- Inside all cells is a semifluid substance called cytosol. 3- All cells contain chromosomes, which carry genes in the form of DNA.

	Prokaryotic cell	Eukaryotic cell
Nucleus النواة	No nucleus. DNA in an unbound region called "nucleoid".	DNA in a nucleus that is bounded by a membranous nuclear envelope.
Organelles العضيات	No membrane-bound organelles	Membrane-bound organelles
Size	Smaller	Larger
Cytoplasm	Bounded by plasma membrane	In the region between the plasma membrane & nucleus
Examples	Bacteria & archaea & cyanobacteria (usually unicellular)	Protists, fungi, animals, and plants, algae (unicellular or multicellular)

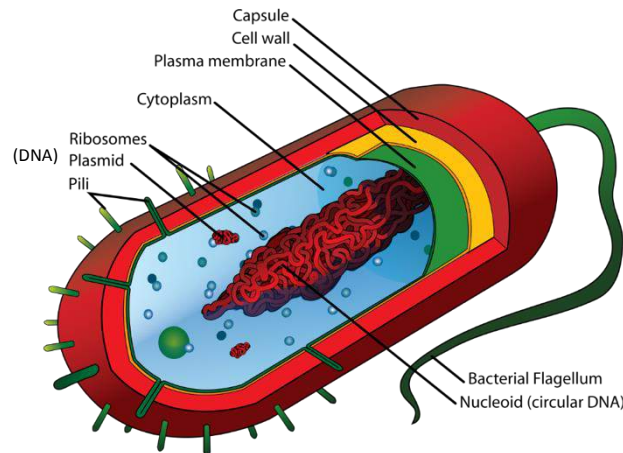
Prokaryotic cells Experiment:

Remember that:

Examples on prokaryotic organisms are bacteria and archaea. These organisms lack membranes and internal cell organelles.



Bacteria slide

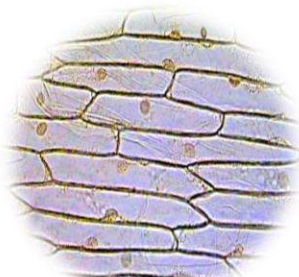


Eukaryotic cells Experiment:

a) Plant cells:

Lugol's solution used to stain plant tissues because its rich with starch (النشا); so Lugol's solution binds to starch therefore the plant tissue appears clearly.

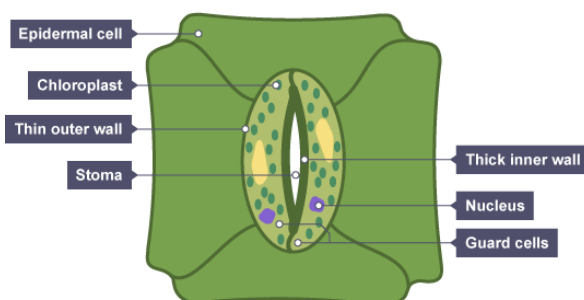
الخطوات كالاتي : قم بسحب طبقة صغيرة من البصل، ضعها على الشريحة، ثم ضع نقطة من الماء H_2O ، ثم ضع نقطة من lugols solution ، ثم قم بتغطية الشريحة و استمتع بمشاهدتها.



Onion epidermis

b) Leaf cells:

الخطوات كالاتي : قم بسحب طبقة صغيرة من ورقة الgeranium، ضعها على الشريحة، ثم ضع نقطة من الماء H_2O ثم قم بتغطية الشريحة ، ضعها تحت المجهر و لا تنسى ان تحت الاجزاء الاتية عليها



Geranium epidermis

Important notes:

- 1- The reason for staining cells is to enhance visualization of the cell and makes colorless parts highlighted. القصد اننا نقوم بصبغ العينات حتى تصبح اوضح ايضا يمكننا من مشاهدة اجزاء عديمة اللون.
- 2- Cell wall (جدار الخلية) composed of cellulose fibers.
- 3- Lugol's iodine solution used to stain plant tissues because its rich with starch(النشا); so Lugol's iodine solution binds to starch therefore the plant tissue appears clearly .
- 4- Chloroplasts (organelles that are important for photosynthesis).

c) Cheek epithelium:

these cells lining the mucus membranes of your mouth and they are easily sloughed off.

They are categorized under "squamous epithelium"

الخطوات كالآتي : قم باستخدام toothpick و حركه في بطانة الفم ، ضع قطرة من الماء على الشريحة، قم بوضع toothpick في قطرة الماء، ثم ضع قطرة من methylene blue كصبغة ، غط الشريحة .



Cheek epithelium

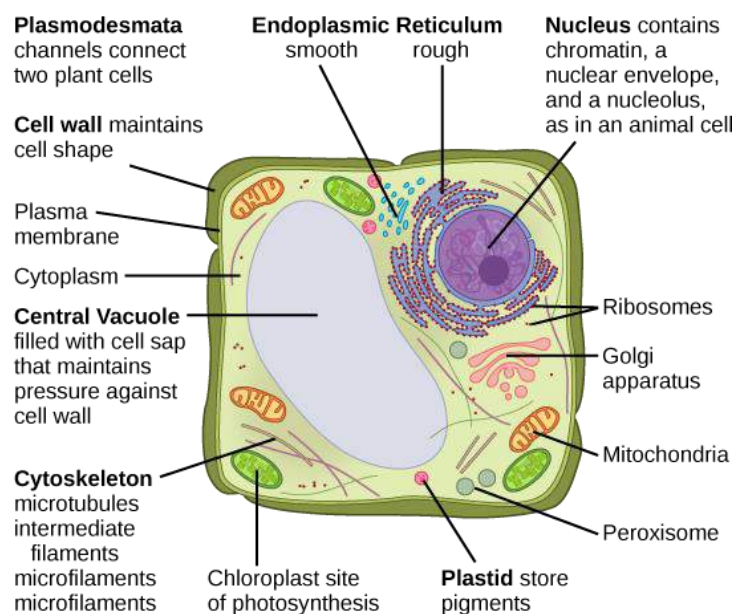
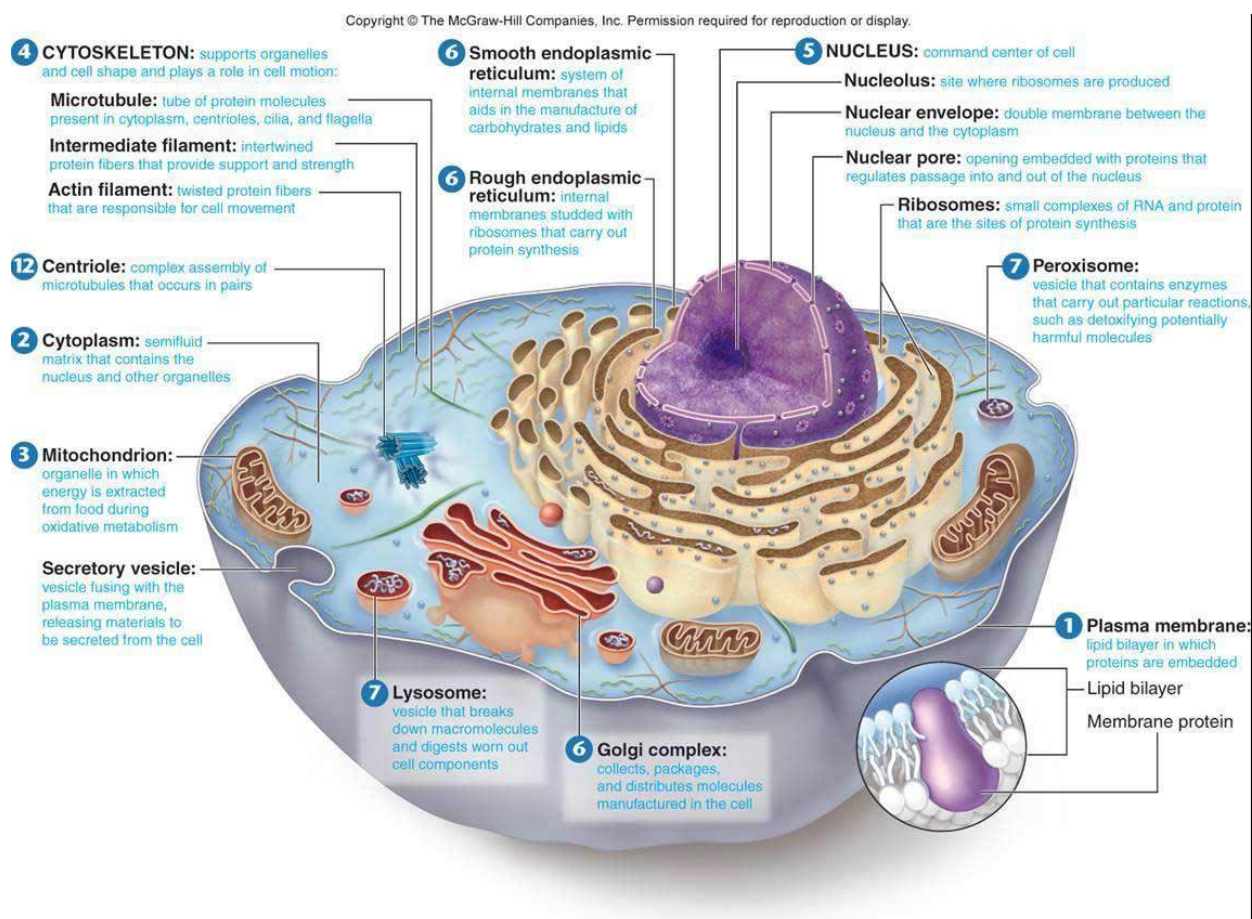
Important notes :

- 1- Methylene blue used to stain animal cells.
- 2- Paramecium is an example of a single-celled (unicellular) eukaryotic cell.

d) Animal and plant cell comparison:

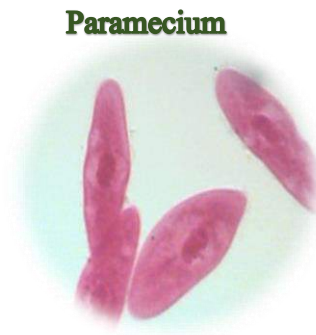
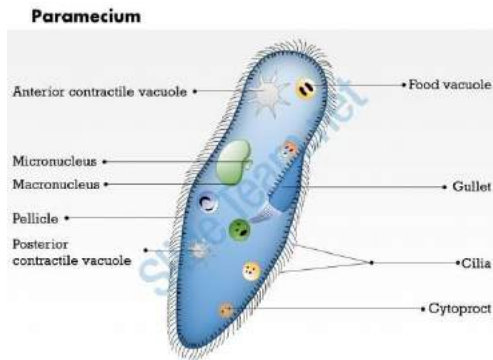
Organelle	Function	Found in
CELL WALL	Maintains cell shape	Plant cells
CELL MEMBRANE	Protects the cell, performs active transport & passive transport, moves materials in and out of the cell, communication	Both
CYTOPLASM	Supports organelles inside the cell.	Both
NUCLEUS	Command center of cell	Both
NUCLEAR ENVELOPE	Covers and protects the nucleus	Both
NUCLEOLUS	Produces ribosome's	Both
ROUGH ENDOPLASMIC RETICULUM	Internal membranes studded with ribosomes that carry out protein synthesis	Both
SMOOTH ENDOPLASMIC RETICULUM	System of internal membranes that aids in manufacture of carbohydrates & lipids	Both
RIBOSOME	Protein synthesis	Both
MITOCHONDRIA	Organelle in which energy is extracted from food during oxidative metabolism	Both
VACUOLE	Filled with cell sap that maintains pressure against cell wall	Both
CHLOROPLAST	Captures sunlight and uses it to produce food through photosynthesis	Plant
GOLGI BODY	Packages and secretes molecules for use in and out of the cell	Both
LYSOSOME	Digests older cell parts, food or other objects	Both
CENTRIOLE	Complex assembly of microtubules that occurs in pairs. Has role in mitosis.	Animal
peroxisome	Site of detoxification إزالة السموم	both

#2 Cell structure & function - 5



e)paramecium:

is an example of single celled eukaryotic organism



1- Why was the cell stained? To enhance visualization of the cell or certain cellular components under a microscope.

2- List five differences between animal and plant cells:

	Animal cells	Plant cells
Cell wall	✗	✓
chloroplast	✗	✓
plastids	✗	✓
Lysosome	✓	✗
Centriole	✓	✗

3- List four differences between Prokaryotic and Eukaryotic cells: see 2nd table.

Test your self

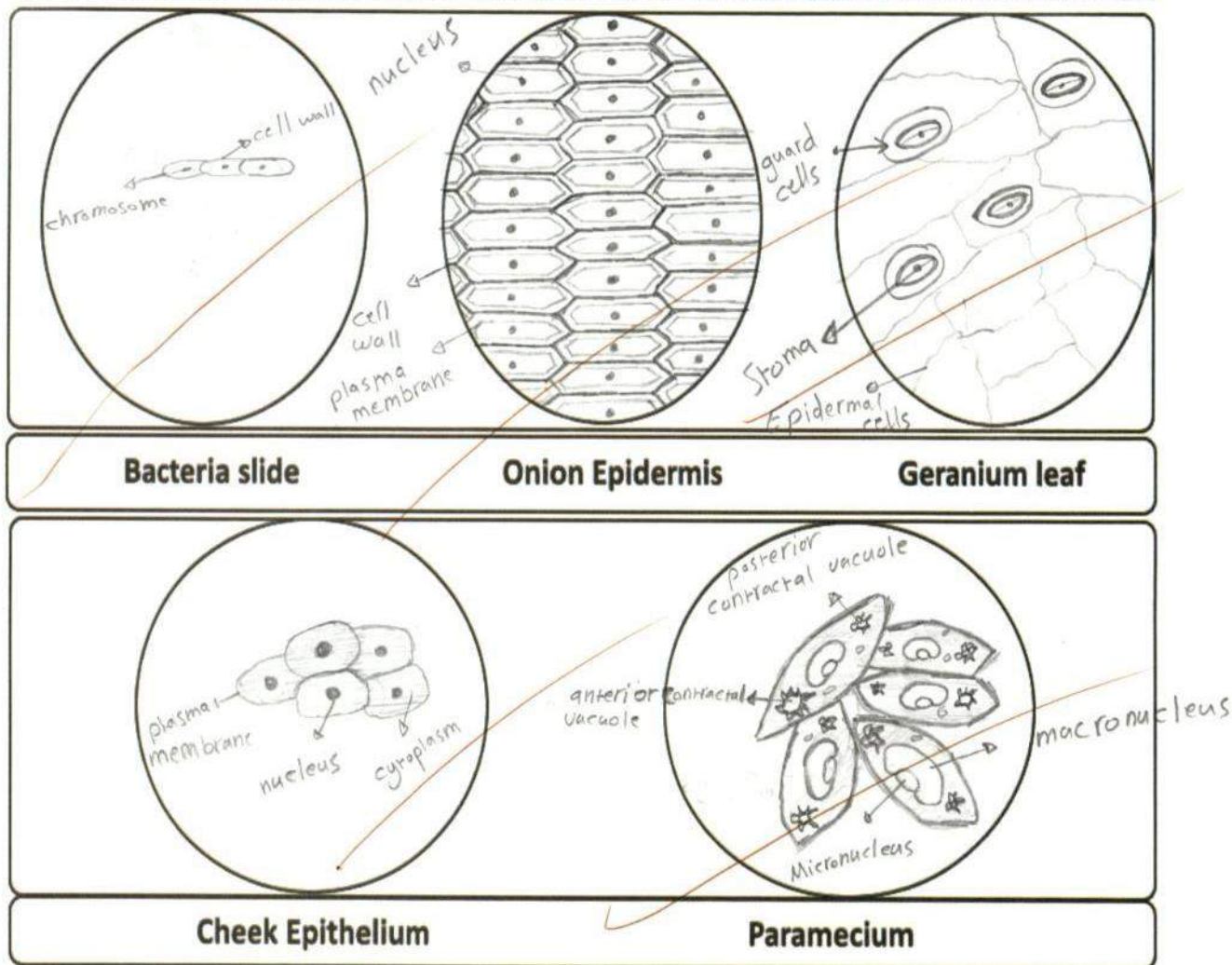
- 1- The cell stained to get high magnification slide
☐ True ☐ False
- 2- Paramecium:
☐ (a) Single-celled prokaryote. ☐ (b) Multicellular eukaryote
☐ (c) Single-celled eukaryote. ☐ (d) Single-celled bacterium.
- 3- Peroxisomes occurs only in animal cells
☐ True ☐ False
- 4- Protein synthesis takes place in
☐ (a) Nucleolus. ☐ (b) Nucleus.
☐ (c) Rough endoplasmic reticulum. ☐ (d) Smooth endoplasmic reticulum.
- 5- Lysosomes occurs only in plant cells.
☐ True ☐ False
- 6- Methylene blue used to stain epidermal cells of onion.
☐ True ☐ False
- 7- Collects, packages and distributes molecules manufactured in the cell:
☐ (a) Lysosome. ☐ (b) Golgi complex.
☐ (c) Peroxisomes. ☐ (d) Nucleus.
- 8- Cytoskeletons are made of cellulose fibers.
☐ True ☐ False
- 9- Guard cells are found in plant cells.
☐ True ☐ False
- 10- Plasmid occurs in.....
☐ (a) plant cell. ☐ (b) paramecium.
☐ (c) amoeba. ☐ (d) bacteria.

Answers

1	2	3	4	5	6	7	8	9	10
False	c	False	a	False	False	b	False	True	d

"ما الماضي إلا حلم، و ما المستقبل إلا رؤية، و عيشك الحاضر بحب تام لله سبحانه وتعالى يجعل من الماضي حلمًا من السعادة ومن المستقبل رؤية من الأمل" ... إبراهيم الفقي

#2 Cell structure & function - 8



1- Why was the cell stained?

To view the specimen because staining highlight structures in biological tissues for viewing and it give the microscopic image some contrast.

2- List five differences between animal and plant cells:

Animal cells	Plant cells
They Don't have vacuoles	They Have central vacuole
They Don't have chloroplasts	They Have Chloroplasts
They Don't have cell wall	They Have cell wall
They Don't have plastids	They Have plastids
The nucleus is in the center of the cell	They The nucleus is not in the center of the cell
They have secretory vesicle	They have plasmodesmata

2- List four differences between Prokaryotic and Eukaryotic cells:

Prokaryotic cells	Eukaryotic cells
No nucleus	Have nucleus
unicellular (Bacteria)	unicellular (paramecium) and multicellular (animal)
Small in size	large in size
No bounded organelles	Have bounded organelles
Cytoplasm bounded with plasma membran	Cytoplasm bounded between plasma membran and nucleus

